# Reliability

*“Conceptual Basis”*

- Nurses attempt to measure the length of babies at birth and at regular intervals thereafter.
- Babies squirm around erratically, and they resist attempts to stretch them out to their full length.
- Such squirming created difficulties for nurses who are attempting to obtain accurate measurements of babies’ length.

- Furthermore;
- Some babies are more compliant than others, which means that some are less likely to squirm around than others.
- This creates differences among babies in that some babies are more likely to squirm around than others.
- And these result in differences among babies in that some babies may be more likely to be measured accurately than others.
- These kind of problems have led researchers to ask questions about the reliability of these measurements.

- Imagine that a nurse is asked to measure the lengths of 10 different babies and imagine that there was some way to know beforehand (but unknown to the nurse) each baby's true length.
- In theory, we could compare each baby's measured length with his or her true length.
- Moreover, we could examine the differences among babies' measured lengths and compare them with the differences among babies' true lengths.

- Ideally, we would find good consistency between these two set of differences.
- That is, we could hope to find that differences in their actual lengths – that the babies were measured (by the nurse) as relatively long were truly were relatively long.
- If this was the case, them we could conclude that the measurement procedure produced length «scores» that were ***reliable.***

- To understand the psychological variability is important.
- Psychological tests are useful only to the degree that they accurately reflect true psychological differences.
- In a research context, behavioral science strives to quantify the degree to which differences in one variable (e.g., intelligence) are associated with differences in other variables (e.g., parenting styles, aggression, academic performance).
- Tests and other types of measures are used to assess and represent these behavioral differences.

- In an applied context, practitioners strive to make decisions about people, and they use behavioral tests and diagnostic procedures to inform those decisions.
- Such decisions rest on the assumption that behavioral differences among people exist, that the differences have important implications, and that they can be measured with some precision.
- Thus, psychological measurement always hinges on the ability to reflect real psychological differences accurately.

- CTT (Classical Test Theory) is a measurement theory that defines the conceptual basis of reliability and outlines procedures for estimating the reliability of psychological measures.
- Suppose that we give a burnout questionnaire to a group of people, and we found that people differ in their scores on the questionnaire.
- We would like to assume that these differences in their questionnaire scores accurately reflect the differences in their ***true levels of burnout***.

- According to CTT, a test's reliability reflects the extent to which the differences in respondents' test scores are a function of their true psychological differences, as opposed to measurement error.
- Sometimes behavioral scientists speak as if reliability is an all-or-none issue.
- For example, someone might ask us whether a particular test is reliable, and we are likely to answer «Yes» or «No». Such an answer seems to treat reliability as if it is a binary issue, with a test either being reliable or unreliable.
- Reliability is, in fact, on a continuum. That is, a procedure for measuring something will be more or less reliable.

# Overview of Reliability and Classical Test Theory

- According to CTT, reliability derives from:
  - Observed scores
  - True scores
  - Measurement error

- **Observed Scores:**
  - The values that are obtained from the measurement of some characteristic of a person
- **True scores:**
  - Real amounts of that characteristic

«The average score that a person would obtain if he or she completed the scale an infinite number of times.»

«The scores that would be obtained if the test or measurement was perfectly precise – that is, if it was unaffected by measurement error.»

«The actual or real level of psychological attribute being measured by a test.»

- Ideally, test users would like to interpret individuals' observed scores as good estimates of their true scores because most behavioral research and decision making are intended to reflect respondents' true psychological characteristics.
- Reliability is the extent to which differences in respondents' observed scores are consistent with differences in their true scores.

- The extent to which «other» characteristics contribute to differences in observed scores is referred to as ***measurement error,*** because they create inconsistency between observed scores and true scores.
- When measuring or quantity of anything, including features of physical objects or psychological characteristics of people, the results of the measurement will be unreliable to some extent.
- There is no such thing as a perfectly reliable measure!

- Errors influence the measurement of psychological attributes.
- Consider what might happen if a class of school children takes a mathematics achievement test.
- Factors other than «math knowledge» are likely to influence children's test performance.
  - Some children who are taking the test might have a cold on the day that take the test
  - Some children might have eaten nutritious breakfast
  - Some may make many lucky guesses on the test
  - Some children might compute the math answers correctly but, by mistake, circle the wrong choice on the answer sheet

- One useful way to think about reliability is in terms of signal and noise – that is, in terms of our ability to detect a signal in the presence of noise.
- In this framework, true scores are the signal that we would like to detect, and measurement error is the noise that is obscuring the signal and making it difficult to detect.
- Reliability can be seen as a ratio of signal and noise:

$$\text{Reliability} = \frac{Signal}{Signal + Noise}$$

- Reliability is the clearest when there is strong signal and/or little noise.

# Observed Scores, True Scores, and Measurement Error

- Reliability depends on two things:
  - The extent to which differences in test scores can be attributed to real inter- or intra-individual differences and
  - The extent to which differences in test scores are a function of measurement error

- In CTT, a person's observed score on a test is a function of that person's true score, plus error.
- If $X_O$ represents an individual's observed test score, if $X_t$ is the individual's true score on psychological characteristics, and if $X_e$ is the amount of error affecting the individual's responses, then we can write the following formula to represent this assumption:

$$X_O = X_t + X_e$$

| Respondent | Observed Score ($X_O$) | | True Score ($X_t$) | | Error ($X_e$) |
|---|---|---|---|---|---|
| **(a) Responses to an original self-esteem questionnaire** | | | | | |
| Ashley | 120 | = | 130 | + | -10 |
| Bob | 145 | = | 120 | + | 25 |
| Carl | 95 | = | 110 | + | -15 |
| Denise | 85 | = | 100 | + | -15 |
| Eric | 115 | = | 90 | + | 25 |
| Felicia | 70 | = | 80 | + | -10 |
| | | | | | |
| Mean | 105.00 | | 105 | | 0 |
| Variance | 608.34 | | 291.67 | | 316.67 |
| SD | 24.66 | | 17.08 | | 17.80 |
| Reliability= $R_{XX}$ = | .48 | $r_{ot} =$ | .69 | $r_{oe} =$ | .72 |
| $r_{te} =$ | .00 | ${r_{ot}}^2 =$ | .48 | ${r_{oe}}^2 =$ | .52 |

* An artificial data set representing six people's responses to a self-esteem questionnaire.

- CTT makes a very important assumption about measurement error.
- It assumes that error occurs as if it is random.
- This means that measurement error is just as likely to inflate any particular score as it is to decrease any particular score.
- We assume that peoples' responses to a psychological test are affected in unpredictable ways that might make their observed scores artificially high or artificially low.
- Across the entire sample of respondents, random measurement error artificially inflates some people's scores, and it artificially deflates other people's scores.

- Because error affects scores as if it random, the inflation and deflation caused by error is independent of the individuals' true levels of self-esteem.
- That is, measurement error can affect someone with a high true level of self-esteem in the same way (and to the same degree) as it affects someone with a high true level of self-esteem.
- If we consider Ashley and Bob, the events that are temporarily affecting their responses have nothing to do with their true level of self-esteem.
- The timing of Ashley's grade on her biology test and the timing of Bob's news about law school are completely unrelated to how high or low their true levels of self-esteem are.

- There are two important consequences of this assumption about error:
  - Error tends to cancel itself out across respondents. That is, error inflates the scores of some respondents and deflates the scores of other respondents in such a way that the average effect of error across respondents is zero.
  - The second apparent randomness of error is that error scores are uncorrelated with true scores. Error affects observed scores in ways that are independent of the respondents’ true levels of self-esteem. Therefore, if we compute the correlation between the individuals'’ true scores and their error scores in the Table.

# Variances in Observed Scores, True Scores, and Error Scores

- Reliability reflects the degree to which differences in observed scores are consistent with differences in true scores.
- Put another way, reliability depends on the links among observed score variability, true score reliability, and error score variability.
- Given the importance of variability for interpreting and evaluating psychological measurement, we need to understand how the first assumption of CTT (i.e., that, for each individual, $X_O = X_t + X_e$) extends to the differences among people.

- How the true differences between people can be obscured by differences in measurement error?

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- Ashley's $X_t$ -Bob's $X_t$ = 130 - 120 = +10
- Ashley's $X_O$ -Bob's $X_O$ = 120 - 145 = -25
- Ashley's true score is higher than Bob's true score, but her observed score is lower than Bob's observed score.

- This inconsistency is created by the measurement error that artificially deflated Ashley's observed score but artificially inflated Bob's observed score.
- This inconsistency means that the apparent 25-point difference between Ashley and Bob is a very poor reflection of the real difference between Ashley and Bob's true levels of self-esteem.

- Because such inconsistencies potentially affect the differences among all the respondents, let us consider the relative variances across all the participants.

$$s_e{}^2 = \frac{\sum(X_e - \overline{X_e})^2}{N}$$

$$s_e{}^2 = \frac{(-10-0)^2+(25-0)^2+(-15-0)^2+(25-0)^2+(-10-0)^2}{6}$$

$$s_e{}^2 = 316.67$$

- This value represents the degree to which error affected different people in different ways.
- High degree of error variance indicates the potential for poor measurement.

- Assuming that an individual's observed score is the sum of individual's true score and error score (i.e., $X_O = X_t + X_e$) and that measurement error is random, it follows that the total variance of the observed scores from a group of individuals equals the sum of their true score and error score variances:

$$s_O{}^2 = s_t{}^2 + s_e{}^2$$

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| Mean | 105.00 | | 105 | | 0 |
| **Variance** | **608.34** | | **291.67** | | **316.67** |
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# Four Ways to Think of Reliability

| | | Conceptual Basis of Reliability: Observed Scores in Relation to …. | |
|---|---|---|---|
| | | **True Scores** | **Measurement Error** |
| **Statistical basis of reliability in terms of …..** | **Proportion of variance** | Reliability is the ratio of true score variance to observed score variance | Reliability is the lack of error variance |
| | | $R_{xx} = \frac{{s_t}^2}{{s_O}^2}$ | $R_{xx} = 1 - \frac{{s_e}^2}{{s_O}^2}$ |
| | **Correlations** | Reliability is the (squared) correlation between observed and true scores | Reliability is the lack of correlation between observed scores and error scores |
| | | $R_{xx} = r_{ot}^2$ | $R_{xx} = 1 - r_{oe}^2$ |

# Reliability as the Ratio of True Score Variance to Observed Score Variance

- Reliability is the proportion of observed score variance that is attributable to true score variance:

$$R_{xx} = \frac{{s_t}^2}{{s_O}^2}$$

$$R_{xx} = \frac{291.67}{608.33} = .48$$

- This value tells us about 48% of the differences that we see among respondents' observed scores can be attributed to differences among their true trait levels.

- The size of the reliability coefficient indicates a test's reliability.
- Reliability ranges between 0 and 1, and larger $R_{xx}$ values indicate greater psychometric quality.
- As $R_{xx}$ increases, a greater proportion of the differences among observed scores can be attributed to differences among true scores.

**What does $R_{xx}$ = 0 mean?**

- We would be much more satisfied with a reliability of .70 or .80 for research purposes and even higher for applied purposes.

| Respondent | Observed Score ($X_O$) | | True Score ($X_t$) | | Error ($X_e$) |
|---|---|---|---|---|---|
| **(b) Responses to revised self-esteem questionnaire** | | | | | |
| Ashley | 135 | = | 130 | + | 5 |
| Bob | 130 | = | 120 | + | 10 |
| Carl | 95 | = | 110 | + | -15 |
| Denise | 85 | = | 100 | + | -15 |
| Eric | 100 | = | 90 | + | 10 |
| Felicia | 85 | = | 80 | + | 5 |
| | | | | | |
| Mean | 105 | | 105 | | 0 |
| **Variance** | **408.33** | | **291.67** | | **116.67** |
| SD | 20.21 | | 17.08 | | 10.80 |
| Reliability= $R_{XX}$ = | .71 | $r_{ot} =$ | .84 | $r_{oe} =$ | .53 |
| $r_{te} =$ | .00 | ${r_{ot}}^2 =$ | .71 | ${r_{oe}}^2 =$ | .29 |

* An artificial data set representing six people's responses to a self-esteem questionnaire.

- After making some revisions, we hope that we will improve the reliability of the questionnaire.
- Individuals' true scores are the same for the revised test as they were in the original test.
- When we observe the differences among the respondents and focus on Ashley and Bob, in the original data there was clear inconsistency between Ashley's and Bob's true score difference and their observed score difference.
- Specifically, Ashley's true score is 10 points higher than Bob's true score, but her observed score was 25 points lower than Bob's observed score.
- This reflects a substantial effect of measurement error.

- In contrast, their observed scores on the revised questionnaire seem to be much more consistent with their true scores.
- Thus, we begin to get a sense that the revised test does a better job of reflecting the true differences among respondents than the original test.
- Reliability for the revised questionnaire:
- $R_{xx} = \frac{291.67}{408.33} = .71$
- For the revised questionnaire, 71% of the observed score variance can be attributed to variance in true scores.

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# Lack of Error Variance

- Error variance represents the degree to which error affects different people in different ways –artificially inflating some people's scores artificially deflating other people's scores.
- These effects obscure the true differences among people (Ashley and Bob)
- Therefore, reliability can be seen as the degree to which error covariance is minimal in comparison with the variance of observed scores.

- Reliability can be seen as the proportion of observed score variance that is attributable to true score variance:

$$R_{xx} = \frac{{s_t}^2}{{s_O}^2}$$

- Observed score variance is the sum of true score variance and error variance:

$${s_O}^2 = {s_t}^2 + {s_e}^2$$

$${s_t}^2 = {s_O}^2 - {s_e}^2$$

$$R_{xx} = \frac{{s_O}^2 - {s_e}^2}{{s_O}^2}$$

$$R_{xx} = \frac{{s_O}^2}{{s_O}^2} - \frac{{s_e}^2}{{s_O}^2} \longrightarrow R_{xx} = 1 - \frac{{s_e}^2}{{s_O}^2}$$

- Note that

$$\frac{{s_e}^2}{{s_O}^2}$$

- Represents the proportion of observed score variance that is a function of error variance.
- Reliability is relatively high when this proportion is relatively small.
- That is, reliability is high when error variance is small comparison with observed score variance.
- For the data from the original self-esteem questionnaire:

$$R_{xx} = 1 - \frac{316.67}{608.33} = 1 - 0.52 = .48$$

- Thus, 52% of the variance in respondents' observed scores on the original questionnaire is produced by measurement error, leaving only 48% attributable to true score differences among the respondents.

# The (Squared) Correlation Between Observed Scores and True Scores

- Reliability is the degree to which differences in observed scores are consistent with differences in true scores.
- Correlation coefficient tells us the degree to which differences in one variable consistent with differences in another variable.
- Thus, reliability can be seen in terms of the (squared) correlation between observed scores and true scores:

$$R_{xx} = {r_{ot}}^2$$

- A reliability of 1.0 would indicate that the differences among respondents' observed scores are perfectly consistent with the differences among their true scores.

# Lack of The (Squared) Correlation Between Observed Scores and True Scores

- Reliability is the degree to which observed scores are uncorrelated with error scores.

$$R_{xx} = 1 - r_{oe}^{2}$$

- Thus, reliability will be relatively strong when observed scores have relatively low correlations with error scores.

# Reliability and the Standard Error of Measurement

- Although the reliability coefficient is an extremely important piece of psychometric information, it does not directly reflect the size of measurement error associated with a test.
- That is, reliability does not tell us, in test score units, the average size of error scores that we can expect to find when a test is administered to a group of people.

Standard deviation of error scores could be a useful way of expressing the amount of error affecting responses to a test.

- 17.80 represents the average size of the absolute values of the error scores.
- 17.80 tells us that, on average, the respondents' observed scores deviated from their true scores by nearly 18 points.
- In contrast, the 10.80 value for the revised questionnaire deviated from their true scores by 11 points.
- Thus, the observed scores on the revised questionnaire are more accurate than the observed scores for the original questionnaire.

| | **Error Score SD** |
|---|---|
| Original Self-esteem Questionnaire | 17.80 |
| Revised Self-esteem Questionnaire | 10.80 |

- The standard deviation of error scores has a special name – ***standard error of measurement.***
- The standard error of measurement represents the average size of the error scores.
- The larger the standard error of measurement, the greater the average difference between the observed and true scores and less reliable the test.

- A test's standard error of measurement is closely related to its reliability.
- We can find reliability ($R_{xx}$) to find the standard error of measurement:

$$se_m = s_o\sqrt{1 - R_{xx}}$$

$s_o$ is the standard deviation of the observed scores.

- Looking at the data from the original questionnaire:

$$se_m = s_o\sqrt{1 - R_{xx}}$$

$$se_m = 24.66\sqrt{1 - .48} = 24.66(.72) = 17.80$$

- This value is exactly equal to the standard deviation that is computed for error scores.